Analog Electronics

- ...ency Response of BJT & MOSFET Amp.
- ...egrated ckt biasing techniques
- M3 • Fundamentals of op Amp. (& Applications)
- M4 • ~~Applications of~~ Power Amp.
- M5 • Oscillators

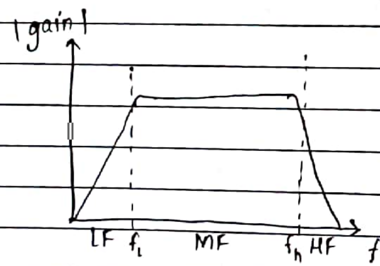
Basic Concepts

* Frequency Response

1) Voltage gain 3) i/p Z
2) Current gain 4) o/p Z

assumed measurements for mid-frequency range.

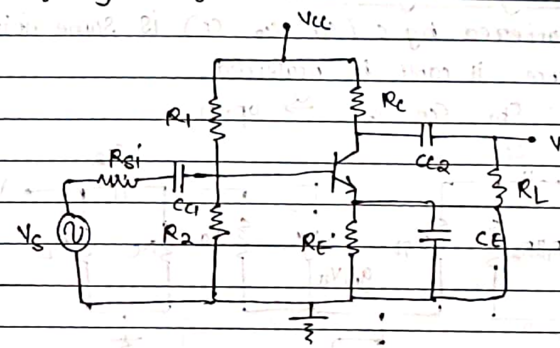
Freq. Ranges : Low, Mid, High

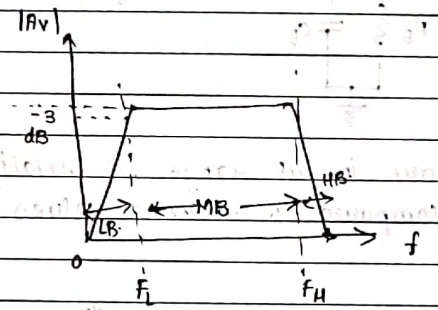


the parameters above are measured only for MF.

Bandwidth = $f_h - f_l$

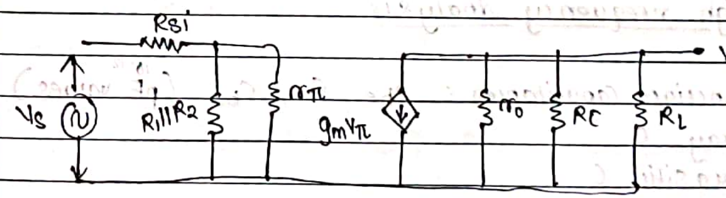
* Frequency Range





- In MF Range : to calculate Av mid (we did AC analysis where all C were shorted), & DC supply was grounded
- C_{C1}, C_{C2} : coupling C
- C_E : Bypass C

• Hybrid π model : (MF)



$$A_V = -g_m (R_L \| r_o \| R_C)$$

9] Effects of c on gain :

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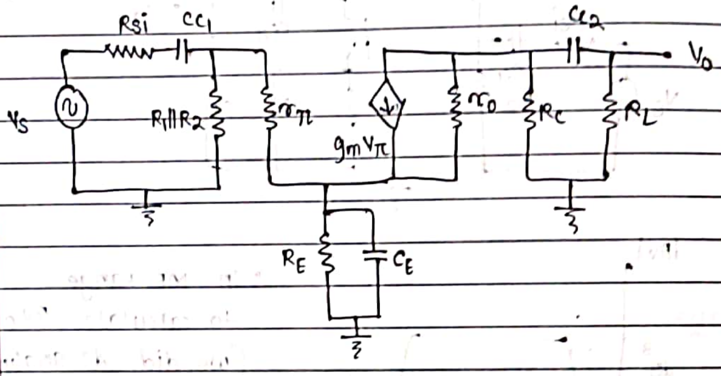
* Low frequency Analysis :

- Reactance offered by c (c_{c1}, c_{c2}, c_E) is some finite value. Hence, it can't be neglected.
- Junction c : c_{be}, c_{ce}, c_{bc} → open



→ c_{c1}, c_{c2} and c_E are in μF ranges, ie. relatively high c-value, compared to other c-values.

and, $x_c = \frac{1}{2\pi f \cdot C}$

→ combined effect of c_{c1}, c_{c2} & c_E reduces gain, ∵ they offer a finite Reactance value to the i/p & o/p signal.

* High frequency Analysis :

- Junction capacitances : c_{be}, c_{ce}, c_{bc} (pF values) 10^{-12}
- Stray c
- Parasitic c

For MF analysis, these junction c assumed to be open bcz x_c tends to ∞.

∴ For HF., x_c offered by these junction c can't be neglected.

- c_{c1}, c_{c2}, c_E → shorted

c_{ce} is very v... small ...

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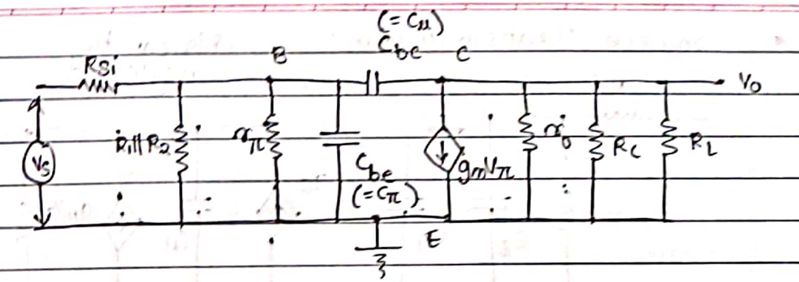
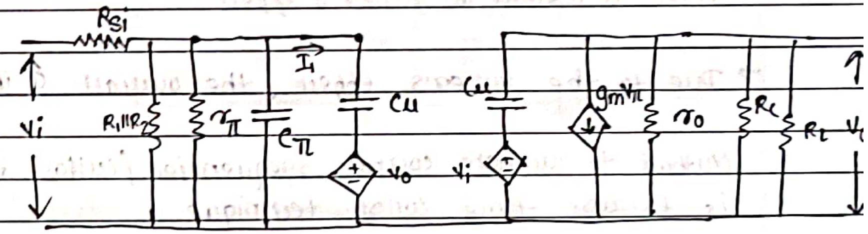


* Miller's Theorem :

- If it is a 2 port Network, then acc. to theorem the element connecting i/p & o/p is divided in 2 parts : a) i/p & ground (connected betn.)

  b) o/p & ground (— " —)



→ $I_1 = \frac{V_\pi - V_0}{1/j\omega C_\mu}$

- here, $V_0 = -g_m V_\pi (r_o)$

∴ $I_1 = j\omega \left[C_\mu \left[1 + g_m (r_o \| R_c) \right] \right] V_\pi$

∴ $I_1 = \frac{V_\pi}{1/j\omega c'}$

- here, $C_m = c' = C_\mu \left[1 + g_m (r_o \| R_c) \right]$ (Miller's capacitance)

∴ $C_m = C_\mu \left[1 + |A_V|_{mid} \right]$

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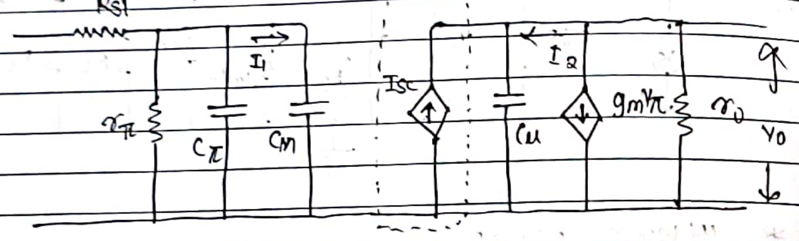
- Source Transformation : this can be neglected



→ Applying Miller's Thm., the C (Cu) is getting multiplied by the term $(1+|A_V|_{mid})$ due to which the overall Cu value is ~~too~~ increased, and this is Miller's capacitance, and the multiplication effect is teumid as Miller's effect.

→ Due to the Miller's effect, the overall C increases

- Method to calculate corner frequencies / cuttoff freq. is to use time const. technique.

$$F = \frac{1}{2\pi\tau}$$

here : τ : time const. of circuit
$(\tau = R_{eq} C_{eq})$

Now,
$$f_{H C\pi} = \frac{1}{2\pi \cdot C_{eq} R_{eq}}$$

$C_{eq} = C_\pi + C_m$

∴ overall C ↑, Higher cutoff freq. ↓ after apply Miller's Thm.

- Due to Miller's effect, Coverall ↑, f_H ↓ ∴ at const. f_L, BW. ↓ ie. Miller's effect reduces Band width.

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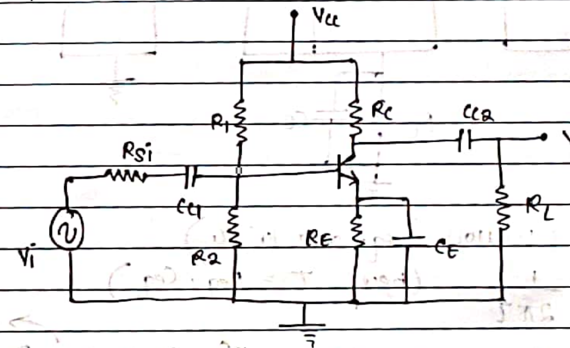
* CE - Bypass

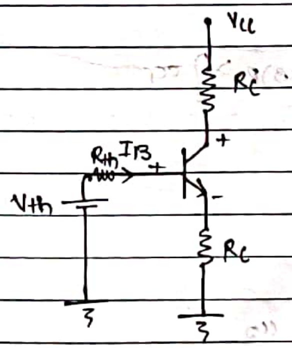
Q] For a CE bypass V-divider bias BJT amp. determine MF. voltage gain, 3 dB f_L, 3 dB f_H and BW., given that C_{C1}, C_{C2} and C_E (μF) along with C_π & C_u (pF)

- For Mid Band Av calculation (DC analysis & AC MF. analysis)
- To calculate f_L (low F. analysis), f_H (HF. analysis)

→ Diagram :



(1) Equivalent DC-Thevenin's ckt and DC Analysis



$$V_{th} = \frac{V_{cc} \cdot R_2}{R_1 + R_2}$$

- Now, KVL on i/p loop :

a) $I_B = \frac{V_{th} - V_{BE}}{R_{th} + (1+\beta) R_E}$

b) $I_C = \beta I_B$

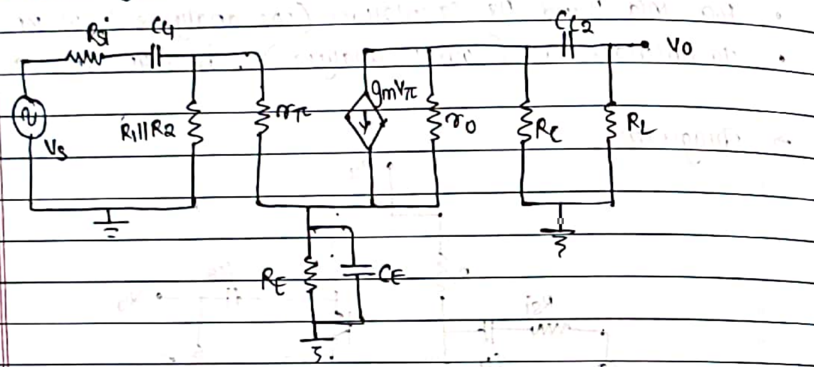
c) calculate V_{CE} only when Q-point

→ Now, $g_m = \frac{I_C}{V_T}$, $r_0 = \frac{V_A}{I_C}$ (If $V_A \neq \infty$), $r_\pi = \frac{\beta V_T}{I_C}$

(2) MF AC analysis

- $A_{V\,mid} = -g_m\,(R_C \parallel R_L \parallel r_o)$

(3) LF Analysis :



a) f_{LC_1} (L-cutoff freq. due to C_1)

$$= \frac{1}{2\pi\tau} \quad (\text{here, } \tau = R_{eq}\cdot C_{eq})$$

Now, $R_{eq} = R_{si} + \left(R_1 \parallel R_2 \parallel [r_\pi + (1+\beta)R_E]\right)$ → UNBYPASS

$C_{eq} = C_{C_1}$ → $= R_{si} + (R_1 \parallel R_2 \parallel r_\pi)$ → BYPASS

$$\therefore f_{LC_1} = \frac{1}{2\pi\left[R_1 \parallel R_2 \parallel (r_\pi + (1+\beta)R_E)\right] C_{C_1}}$$

b) f_{LC_2} :

$R_{eq} = R_C + R_L$

$C_{eq} = C_{C_2}$

$$\therefore f_{LC_2} = \frac{1}{2\pi[\cancel{R_{si}}\ R_C + R_L]\,C_{C_2}}$$

c) f_{LCE} :

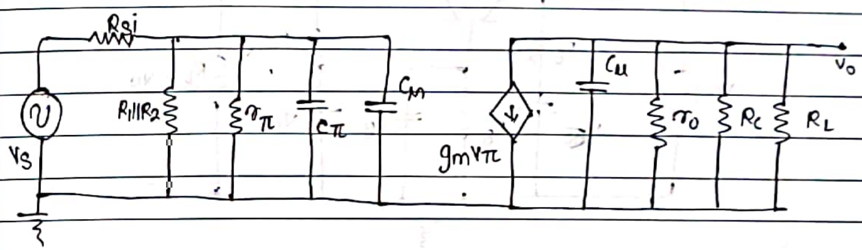
$$R_{eq} = R_E \parallel \frac{(R_{si} \parallel R_1 \parallel R_2 + r_\pi)}{1+\beta}$$

$C_{eq} = C_E$

$$\therefore f_{LCE} = \frac{1}{2\pi\left[R_E \parallel \frac{(R_{si} \parallel R_1 \parallel R_2 + r_\pi)}{1+\beta}\right] C_E}$$

(Largest of 3 is f_L)

(4) HF Analysis :



a) $$f_{Hi} = \frac{1}{2\pi[R_{si} \parallel R_1 \parallel R_2 \parallel r_\pi](C_\pi + C_m)}$$

here,

$C_m = C_\mu\,[1 + |A_V|_{mid}]$

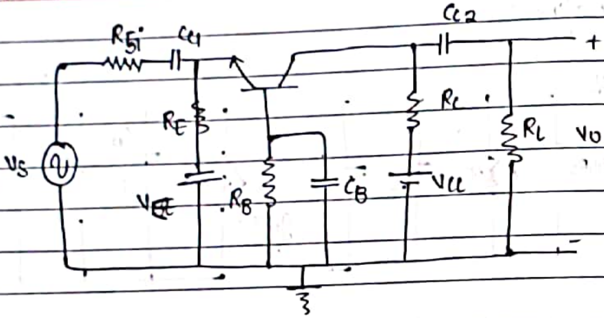
b) $$f_{Ho} = \frac{1}{2\pi[r_o \parallel R_C \parallel R_L](C_\mu)}$$

(Smallest of 2 is f_H)

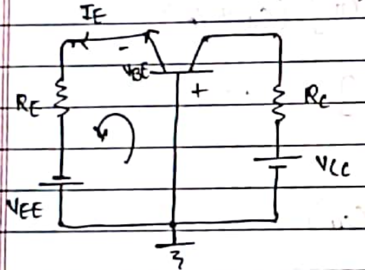
→ Now,

$\boxed{BW = f_H - f_L}$

* CB - BJT Amp :



→ DC Analysis.



• KVL on i/p side

$\therefore -V_{BE} - I_E R_E + V_{EE} = 0$

$\therefore I_E = \dfrac{V_{CC} - V_{BE}}{R_E}$

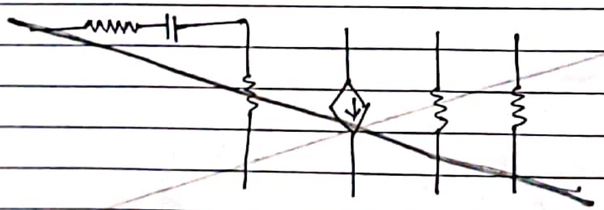
Now,

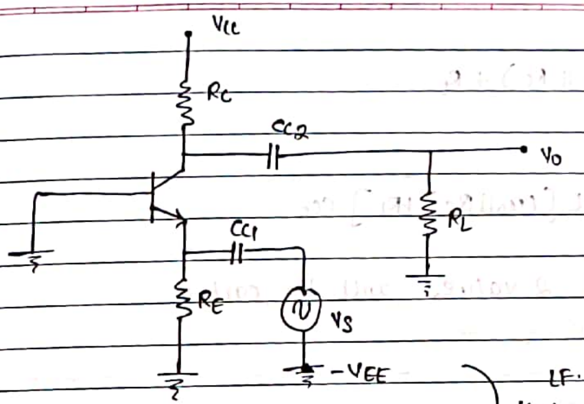
$I_B = \dfrac{I_E}{1+\beta}$

$I_C = \beta I_B$

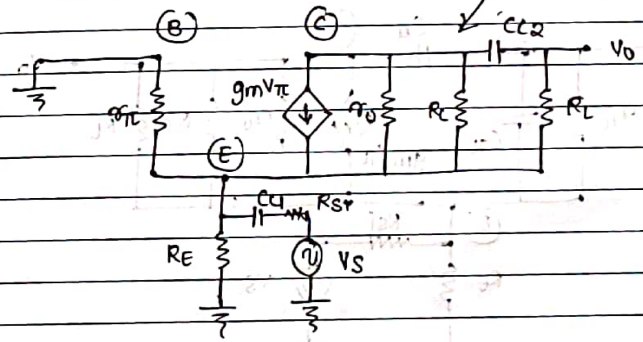
$$\boxed{\begin{array}{l} \text{and} \rightarrow g_m = \dfrac{I_C}{V_T} \quad \rightarrow r_\pi = \dfrac{\beta V_T}{I_C} \\ \rightarrow r_o = \dfrac{V_A}{I_C} \end{array}}$$

→ LF. Analysis :





LF.
Hybrid π model



a) $f_{LC1} = \dfrac{1}{2\pi\tau}$

$R_{eq} = R_{Si} + \left[R_E \parallel \dfrac{r_\pi}{1+\beta} \right]$

$C_{eq} = C_{C1}$

$\therefore f_{LC1} = \dfrac{1}{2\pi \left[R_{Si} + \left(R_E \parallel \dfrac{r_\pi}{(1+\beta)} \right) \right] C_{C1}}$

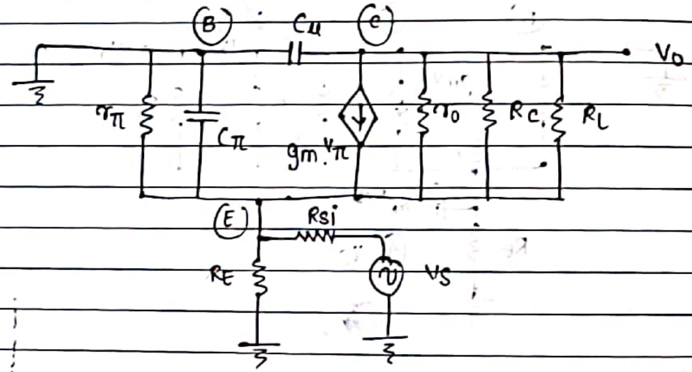
b) f_{LU_2} :

$R_{eq} = (r_o \parallel R_C) + R_L$

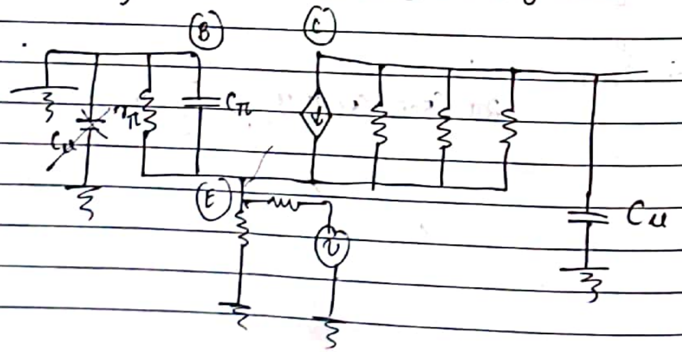
$\therefore f_{LU_2} = \dfrac{1}{2\pi\,[(r_o \parallel R_C) + R_L]\, C_{C_2}}$

- Largest of 2 values will be called as 3dB f_L.

→ HF Analysis :



- Applying Miller's Thm. :
Replacing the Cu (cap.) betn. Base & C
with i) Cu (cap) betn. B and ground.
ii) Cu (cap) betn. C and ground.



- at the i/p side Cu is connected betn. B and ground but B itself is grounded,
$\therefore$ Cu becomes redundant, $\therefore$ Cu is to be neglected.

This signifies,
Hence there is no Miller effect, $\therefore$ Ceq decreases and hence, BW ↑.

→ This concludes, BW offered in CB > BW in CE config.
Reason: No Miller's effect.

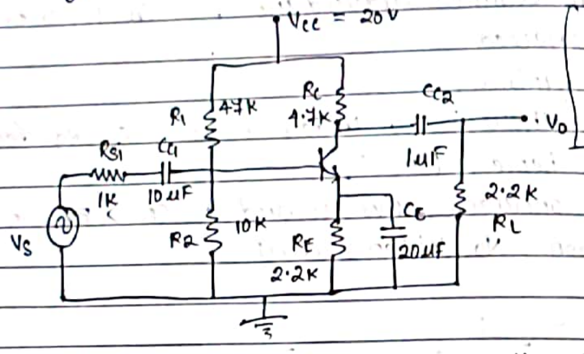
- $f_{Hi} = \dfrac{1}{2\pi \left[R_{si} \parallel R_E \parallel \dfrac{r_\pi}{1+\beta} \right] C_\pi}$

- $f_{Ho} = \dfrac{1}{2\pi\, [r_o \parallel R_C \parallel R_L] \cdot C_\mu}$

→ Mid freq. Analysis

$|A_v|_{mid} = g_m (r_o \parallel R_C \parallel R_L)$

9] Determine the upper, lower f_c, calculate BW. for the ckt. given, also calculate Gain - BW. product.



$C_\pi = 35pF$

$C_u = 5pF$

$r_o = \infty$

→ Solution :

A) DC Analysis :

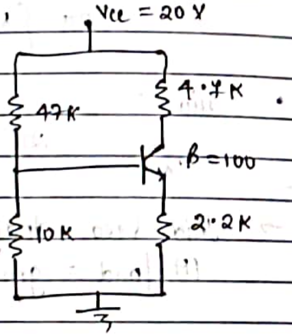
- Now,

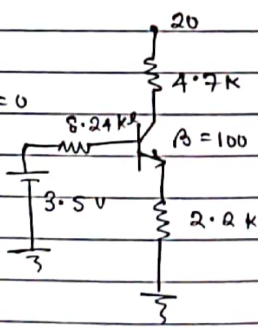
$$R_{th} = \frac{(R_1)(R_2)}{R_1 + R_2} = \frac{(47)(10)(10^6)}{57(10^3)}$$

$$= 8.24\ k\Omega$$

$$V_{th} = \frac{V_{cc}(R_2)}{R_1 + R_2} = \frac{(20)(10)(10^3)}{57(10^3)}$$

$$= 3.5\ V$$





- $V_{th} - I_B R_{th} - V_{BE} - R_E \cdot I_E = 0$

$\therefore\ 3.5 - I_B(8.24)(10^3)$

$- 0.7 - (2.2)(10^3)(100+1) I_B = 0$

$\therefore I_B = 12.2\ \mu A$

$\therefore I_C = 1.22\ mA = \beta I_B$

- $$g_m = \frac{I_C}{V_T} = \frac{1.22(10^{-3})}{26(10^{-3})}$$

$$= 46.9\ m℧$$

- $$r_\pi = \frac{\beta V_T}{I_C} = \frac{(100)(26)(10^{-3})}{1.22(10^{-3})}$$

$$= 2131.14\ \Omega$$

$$= 2.13\ k\Omega$$

- $r_o = \infty$

B) AC Analysis :

- $A_{v\,mid} = -g_m [R_C \parallel R_L \parallel r_o]$

$= -(46.9)(10^{-3})\,[4.7(10^3) \parallel 2.2(10^3)]$

$= -70.2$

- c) ~~LF Analy.~~ $A_{vs\,mid} = -g_m (R_C \parallel R_L \parallel r_o) \left[\frac{R_B \parallel r_\pi}{R_B \parallel r_\pi + R_{si}} \right]$

$= -(46.9)(10^{-3})[1.49](10^3) \left[\frac{1.69(10^3)}{\cancel{1001.69}\ 2.69(10^3)} \right]$

$= -44.10$

c) LF Analysis :

a) $$f_{LL1} = \frac{1}{2\pi\,[r_\pi \parallel R_1 \parallel R_2 + R_{si}]\,(10)(10^{-6})}$$

$$= \frac{1}{2\pi([2.13K \parallel 8.24K] + 1000](10)(10^{-6})}$$

$$= \frac{1}{2\pi(2.69)(10)(10^{-6})(10^3)}$$

$$= 5.9\ Hz$$

b) $$f_{LL2} = \frac{1}{2\pi(R_C + R_L)\,C_{C2}} = 23\ Hz$$

c) $$f_{LE} = \frac{1}{2\pi \left[R_E \parallel \left(\frac{R_{si} + r_\pi \parallel R_1 \parallel R_2}{1+\beta} \right) \right]} = 265\ Hz$$

$$= \frac{1}{2\pi \left[2.2(10^3) \parallel \left(2.13(10^3) + \frac{1(K) \parallel 47K \parallel 10K}{101} \right) \right]}$$

147 KHz, 7.5 MHz

D) HF Analysis:

a) $$f_{Hi} = \frac{1}{2\pi (r_\pi \parallel R_1 \parallel R_2 \parallel R_{si})(C_\pi + C_m)}$$

here, $C_m = C_\mu (1 + A_{vs\,mid})$

$\therefore C_m = 5(10^{-12})[-43.1]$

$= 215.5 \text{ pF}$

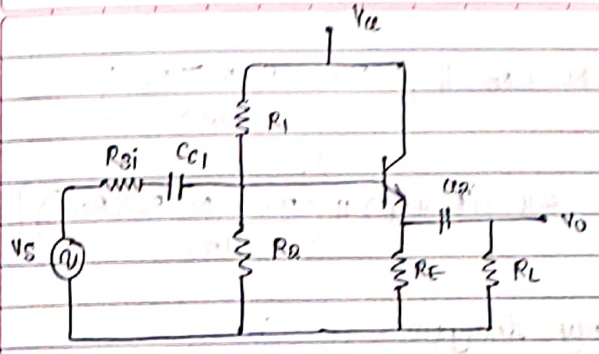
$$\therefore f_{Hi} = \frac{1}{2\pi [2.13K \parallel 47K \parallel 10K \parallel 1K](215.5)(10^{-12})}$$

$$= \frac{1}{2\pi [2.03K \parallel 0.909K][(215.5)(10^{-12}) + 35]}$$

$$= \frac{1}{2\pi (0.628)(10^3)(215.5 + 35)(10^{-12})}$$

=

CC - BJT Amplifier

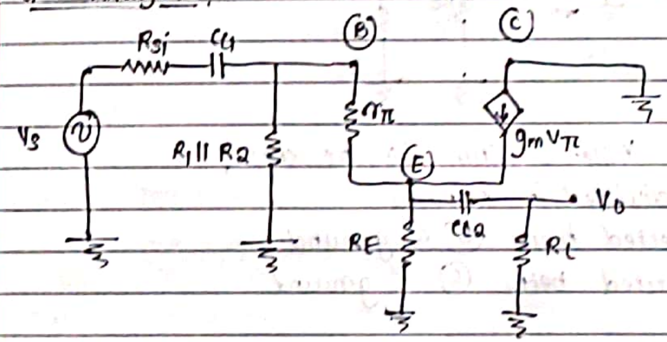


- DC Analysis is same as CE-BJT amp.

→ MF AC Analysis:

$$A_v = \frac{(1+\beta)(r_o \parallel R_E)}{r_\pi + (1+\beta)(r_o \parallel R_E)} \approx 1$$

→ LF Analysis:



- $$f_{Lc1} = \frac{1}{2\pi \cdot R_{eq} \cdot (C_{c1})}$$

Now, $R_{eq} = R_{si} + [R_1 \parallel R_2 \parallel (r_\pi + (R_E \parallel R_L)(1+\beta))]$

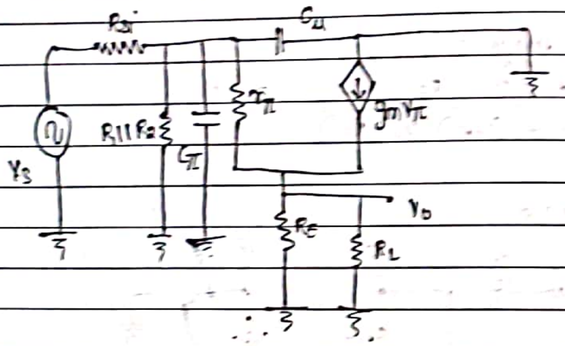
$$\therefore f_{Lc1} = \frac{1}{2\pi [R_{si} + (R_1 \parallel R_2 \parallel (r_\pi + (R_E \parallel R_L)(1+\beta)))] C_{c1}}$$

- f_{H2} :

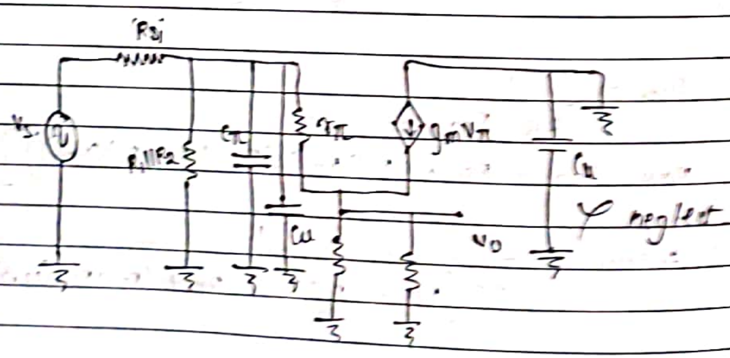
$$R_{eq} = R_L + R_E \,\|\, \left[\frac{r_\pi + R_1 \| R_2 \| R_{si}}{(1+\beta)} \right]$$

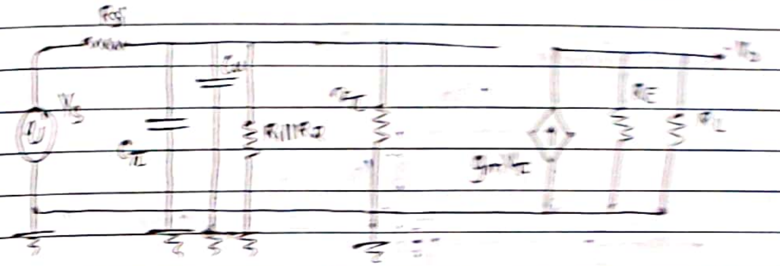
$$\therefore f_{H2} = \frac{1}{2\pi \left[(R_L + R_E') \,\|\, \left(\frac{r_\pi + R_1 \| R_2 \| R_{si}}{(1+\beta)} \right) \right] C_2}$$

→ High Frequency Analysis :



- Applying Miller's Thm. to the ckt.:
  C_μ is divided in 2 parts:
a) C_μ connected betn. (B) & ground.
b) C_μ connected betn. (C) & ground





- $$f_{H1} = \frac{1}{2\pi \left[R_{si} \| R_1 \| R_2 \| r_\pi (1+g_m R') \right] \left[C_\mu + \frac{C_\pi}{1+g_m R'} \right]}$$

→ CC - configuration :

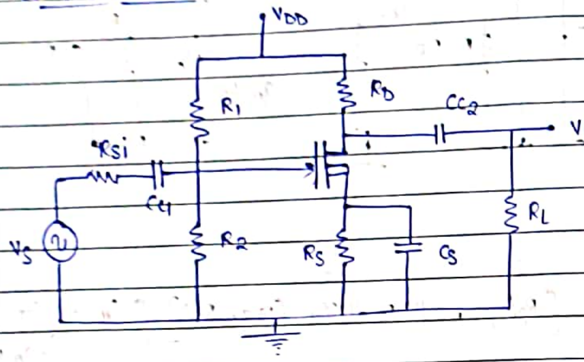
- There is no Miller effect at the output and the capacitance at i/p side (C_π) is reduced by $[1+g_m R']$, so overall C_{eq} is less as compared to CE-config. Hence, we say BW of CC-config is largest of the three.
- No Miller's effect on i/p & o/p side.

→ Gain : (CE = CB) (only phase is diff), (CC - unity)

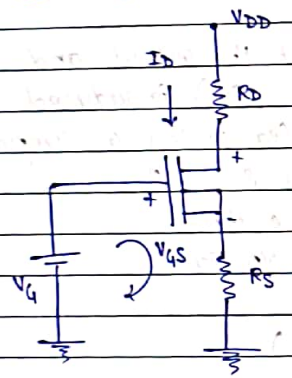
BW : CE < CB < CC

MOSFET Amplifier Frequency Response

A) Common Source :



* DC Analysis



- $V_G = \dfrac{V_{DD} \cdot R_2}{R_1 + R_2}$
- NOW,

$V_G - V_{GS} - I_D R_S = 0$

$\therefore V_G = V_{GS} + I_D R_S$

- NOW,

$I_D = K_n [V_{GS} - V_{TN}]^2$

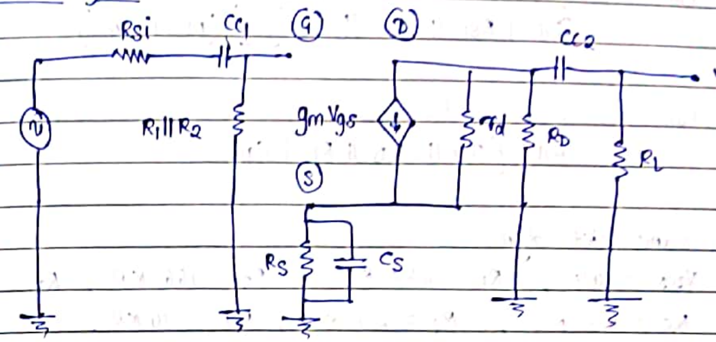
* Parameters :

$g_m = 2K_n [V_{GS} - V_{TN}]$

* Mid-frequency Analysis :

- $(A_v)_{mid} = -g_m [r_d \parallel R_D \parallel R_L]$
- $(A_{vs})_{mid} = -g_m (r_d \parallel R_D \parallel R_L] \dfrac{(R_1 \parallel R_2)}{(R_1 \parallel R_2) + R_{si}}$

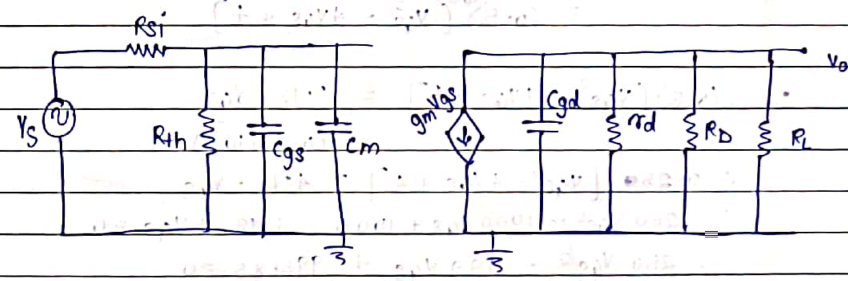
* LF Analysis :



- $f_{LC1} = \dfrac{1}{2\pi [R_{si} + R_{th}] C_{C1}}$
- $f_{LC2} = \dfrac{1}{2\pi [R_D + R_L] C_{C2}}$
- $f_{LS} = \dfrac{1}{2\pi [R_S \parallel 1/g_m] C_S}$; $\because 1/g_m \ll (R_D \parallel R_1)$

→ R. offered by $(g_m V_{gs})$ Source is always $= 1/g_m$

* HF Analysis :



here, $C_m = C_{gd} [1 + |A_v|_{mid}]$

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- $f_{Hi} = \dfrac{1}{2\pi\,[R_{si} \parallel R_1 \parallel R_2]\,[C_{gs} + C_m]}$

- $f_{Ho} = \dfrac{1}{2\pi \cdot [r_d \parallel R_D \parallel R_L]\, C_{gd}}$

Q] Same ckt dgm.

- $V_{DD} = 10\,V$, $R_1 = 234\,k\Omega$, $R_2 = 166\,k\Omega$, $R_D = 4\,k\Omega$, $R_S = 0.5\,k\Omega$, $R_L = 20\,k\Omega$; $R_{si} = 10\,k\Omega$.
- $K_n = 0.5\,mA/V^2$, $V_{TN} = 2\,V$, $\lambda = 0$, $C_{gs} = 1\,pF$, $C_{gd} = 0.1\,pF$, $C_{C1} = C_{C2} = 1\,\mu F$, $C_S = 10\,\mu F$

Calculate BW of CS amplifier.

→ Solution

* DC Analysis:

- $V_G = \dfrac{10(166)(10^3)}{(234+166)\,K} = 4.15\,V$

- Now,

$I_D = \dfrac{V_G - V_{GS}}{R_S} = \dfrac{4.15 - V_{GS}}{0.5(10^3)}$

also, $I_D = (0.5)(10^{-3})[V_{GS} - 2]^2$

$= (0.5)(10^{-3})[V_{GS}^2 - 4V_{GS} + 4]$

- $(0.5)[V_{GS}^2 - 4V_{GS} + 4] = \dfrac{4.15 - V_{GS}}{(0.5)(10^3)}$

$\therefore 0.25\,[V_{GS}^2 - 4V_{GS} + 4] = 4.15 - V_{GS}$

$\therefore 250\,V_{GS}^2 - 1000\,V_{GS} + 1000 - 4.15 + V_{GS} = 0$

$\therefore 250\,V_{GS}^2 - 999\,V_{GS} + 995.85 = 0$

$V_{GS} = 2.1\,V,\ 1.9\,V$

- when, $V_{GS} = 2.1$

$I_D = (0.5)(0.1)^2$

$= 5\,mA$

$V_{GS} = 1.9$

$I_D = (0.5)(-0.1)^2$

$= 5\,mA$

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$\therefore 0.25\,V_{GS}^2 - V_{GS} + 1 = 4.15 - V_{GS}$

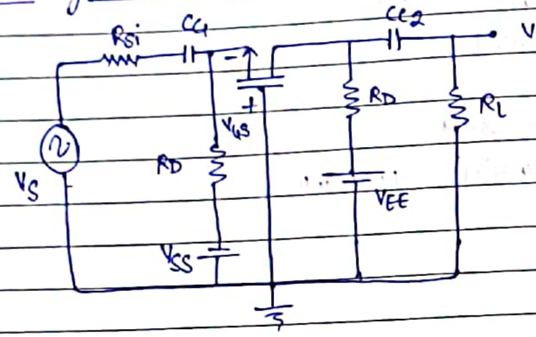
$\therefore 0.25\,V_{GS}^2 - 3.15 = 0$

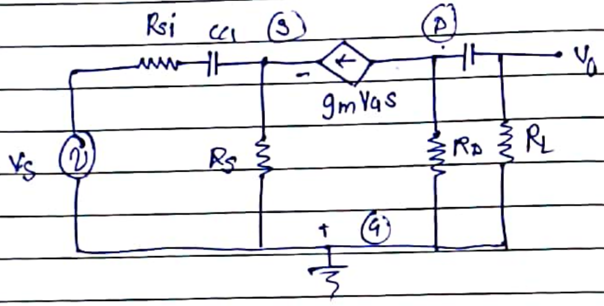
$\therefore V_{GS} = \pm 3.54$

- when, $V_{GS} = 3.54$

$\therefore I_D = \cancel{(0.5)(} \quad 1.18\,mA$

* **Common Gate MOSFET Amplifier :**

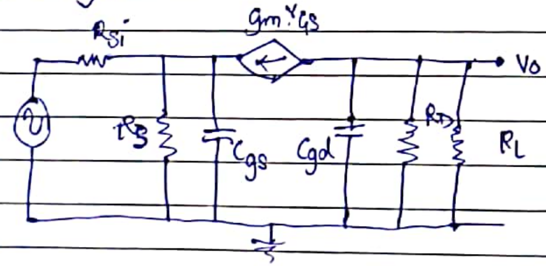




- $f_{LC1} = \dfrac{1}{2\pi\left[R_{si} + (R_S \parallel 1/g_m)\right] C_{C1}}$

- $f_{LC2} = \dfrac{1}{2\pi\left[R_D + R_L\right] C_{C2}}$

→ **HF Analysis :**



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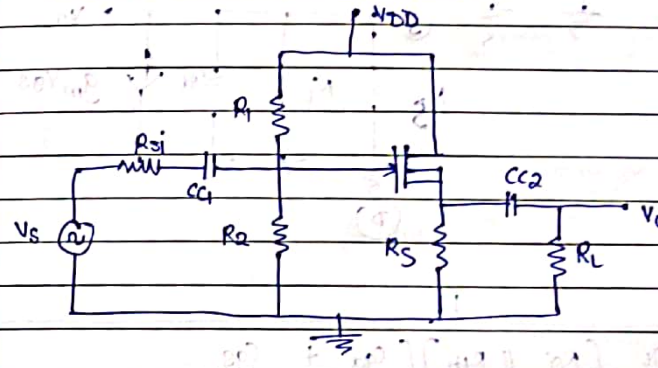
- $f_{Hi} = \dfrac{1}{2\pi\left[R_{si} \parallel R_S \parallel \frac{1}{g_m}\right] C_{gs}}$

- $f_{HO} = \dfrac{1}{2\pi\left[R_D \parallel R_L\right] C_{gd}}$

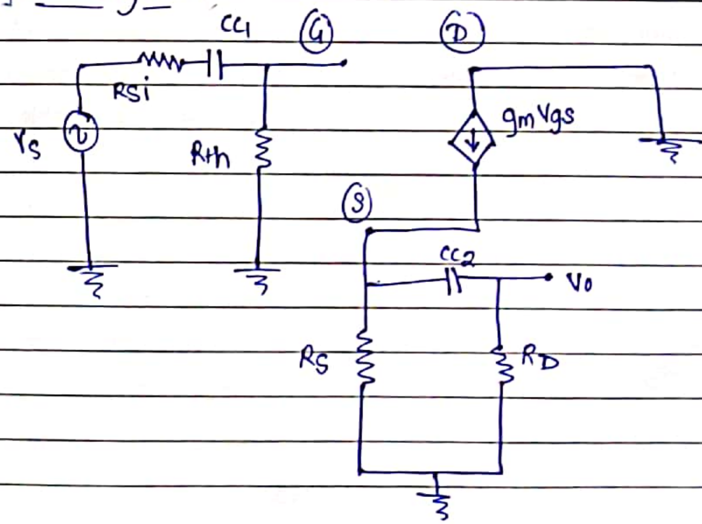
→ $A_v(mid) = +g_m(R_D \parallel R_L)$

$A_{vs}(mid) = \dfrac{g_m(R_D \parallel R_L)}{1 + g_m R_{si}}$

* **Common Drain MOSFET Amplifier :**

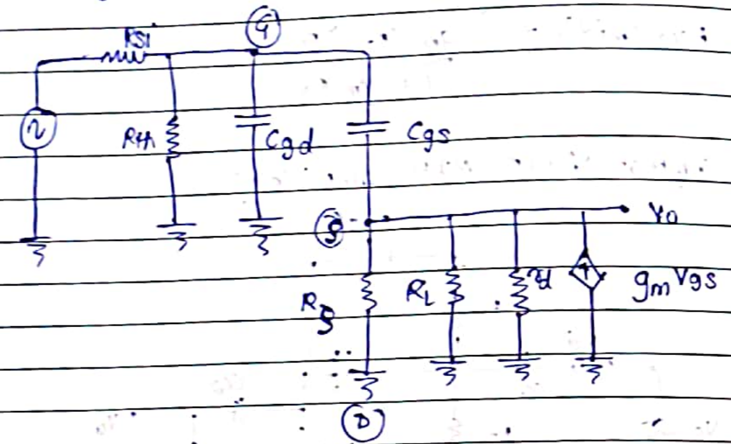


→ **LF Analysis :**



- $f_{LL_1} = \dfrac{1}{2\pi\,[R_{Si} + R_{th}]\,C_{C_1}}$
- $f_{LL_2} = \dfrac{1}{2\pi\,[R_L + (R_S \,\|\, r_d)]\,C_{C_2}}$

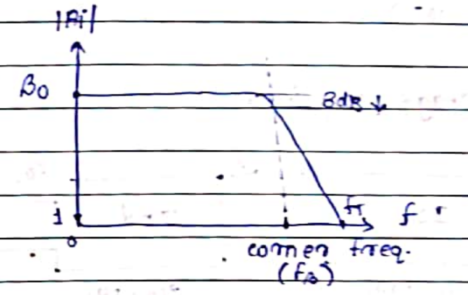
→ HF Analysis:



- $f_{H_1} = \dfrac{1}{2\pi\,[R_{Si} \,\|\, R_{th}]\left[C_{gd} + \dfrac{C_{gs}}{1 + g_m(R_S \,\|\, r_d \,\|\, R_L)}\right]}$
- f_{H_2}

* Short ckt. i gain of BJT Transistor:

- $A_i = \dfrac{g_m r_\pi}{1 + wj\, r_\pi (C_\pi\, C_\mu)}$ → (HF SC i gain)
- $A_i = g_m r_\pi = \beta_0$ → (LF i gain)
- HF i gain is the func. of frequency whereas LF i gain is independen of freq.



- $f_\beta = \dfrac{1}{2\pi \cdot r_\pi (C_\pi + C_\mu)}$
- Unity gain BW. (f_T) (i.e. when $\beta_0 = 1$)
  $\therefore f_T = \beta_0 \cdot f_\beta$
- gain ↓ BW ↑
  gain ↑ BW ↓ } → f_T (gain, BW product)

Q] In certain BJT amp., $r_\pi = 2\,k\Omega$, $\beta = 100$ at 1 MHz, $\beta = 5$ at 20 MHz, determine the value of f_T, f_β and C_π.
Assume: $C_\mu = 1\,pF$.

→ Soln.:

- $f_T = \beta f_\beta$

→ when, $\beta = 100$
$f_T = (100)(10^6)$
$= 10^8 = 100\,MHz$

→ When, $\beta = 5$
$f_T = (5)(20)(10^6)$
$= 100\,MHz$

(BJT) (MOSFET)

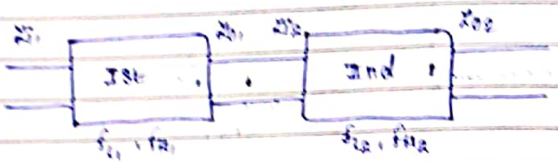
- $f_T = \frac{1}{2\pi\, r_\pi (C_\pi + C_\mu)} = \frac{g_m}{2\pi (C_{gs} + C_{gd})}$

$\therefore (10^6) = \frac{1}{2\pi (2)(10^3)(C_\pi + 10^{-12})}$

$C_\pi + 10^{-12} = \frac{1}{4\pi (10^3)(10^6)} = \frac{10^{-9}}{4\pi}$

$\therefore C_\pi = 3.9\ (10^{-})$

* <u>CS-CS Multistage Amplifier</u> :



- $f_{LT} = \frac{f_L}{\sqrt{2^{1/n} - 1}}$, n: stages
- $f_{HT} = f_H \left(\sqrt{2^{1/n} - 1}\right)$

Q] Assume a 2 stage amp, each stage has: $f_L = 800$ Hz, $f_H = 1000$ KHz. Calculate BW of 2 stage amp.

- $f_{LT} = \frac{(800)}{\sqrt{2^{1/2} - 1}} = 466.25$ HK

- $A_{VT} = A_{V_1} \times A_{V_2} \times \cdots A_{V_n}$

- $f_{HT} = (10^6)\sqrt{2^{1/2} - 1} = 643.59$ KHz

- $\therefore BW = f_{HT} - f_{LT}$
  $= 642.96$ KHz

- BW (single stage) $= 999.2$ KHz (calculated by taking n=1)

- In multistage amp, we get more gain, but compromising with BW

→ MS. amp types :- 1) Cascade amp.
2) Cascode amp.

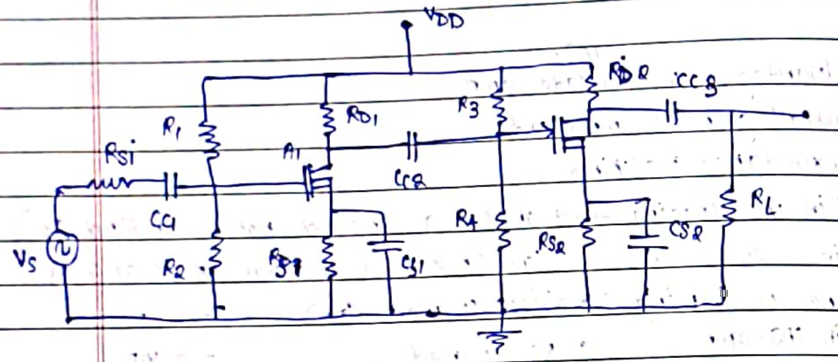
→ Advantages of MS. cascade amp.
- Overall gain is improved:
- Impedance matching
(in MS., loading effect of 1st g, should not be on 2nd g
ie. Z_{o1} & Z_{i2} (low o/p z of 1 & low i/p z of 2)

note
- types of coupling in MS. amp. :
  - direct coupled (DC) → LF.
  - RC coupled → MF
  - transformer coupled → HF.

- RC coupled MS. amp is used, used to isolate the 2 stages using various techniques.

→ Disadvantage :
- compromization with BW.
- Ckt. is more complex, as stages ↑; complexity ↑.

* Cascode CS-CS Multistage MOSFET Amplifier

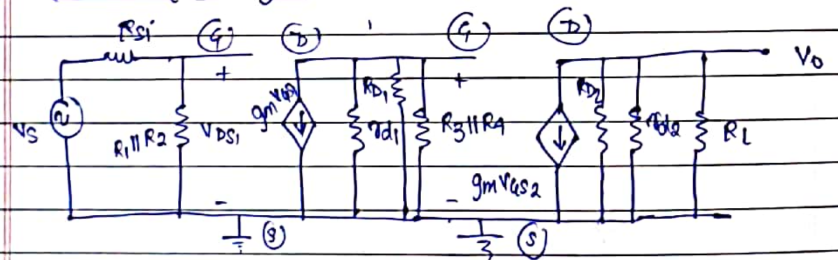


A] DC Analysis:

- $V_{TH_1} = V_{G_1} = \dfrac{V_{DD}(R_2)}{R_1 + R_2}$
- $V_{GS_1} = V_G - I_{D_1} R_{S_1}$

  $I_{D_1} = K_{n_1}\left[V_{GS_1} - V_{TN_1}\right]^2$
- $V_{DS_1} = V_{DD} - I_D(R_{D_1} + R_{S_1})$
- $g_{m_1} = 2K_{n_1}\left[V_{GS_1} - V_{TN_1}\right]$

B] Mid Freq. Analysis:



- Now,

  $A_{V_1} = -g_{m_1}\left[r_{d_1} \| R_3 \| R_4 \right] \| R_{D_1}]$

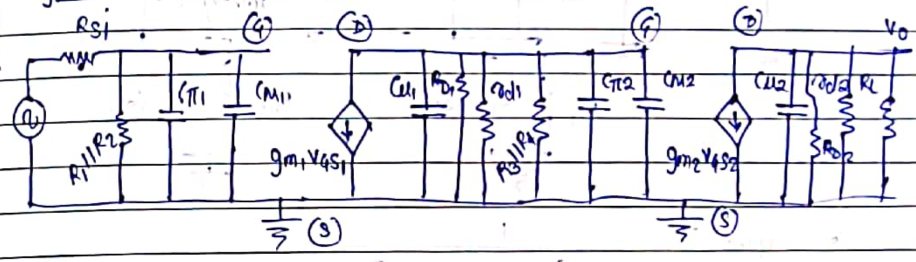
  $A_{V_2} = -g_{m_2}\left[r_{d_2} \| R_L\right]$

  $\therefore A_{V(mid)} = |A_{V_1}|\,|A_{V_2}|$

- $Z_{i_1} = R_{Si} + (R_1 \| R_2)$

  $Z_{o_1} = Z_{i_2} = r_{d_1} \| R_3 \| R_4$
- $Z_{o_2} = r_{d_2} \| R_L$

C] High Frequency Analysis:



- $f_{Hi_1} = \dfrac{1}{2\pi \left[R_{Si} \| R_1 \| R_2\right](C_{\pi 1} + C_{M_1})}$

  here,

  $C_{M_1} = C_{u_1}\left[1 + |A_{V_1}|_{mid}\right]$

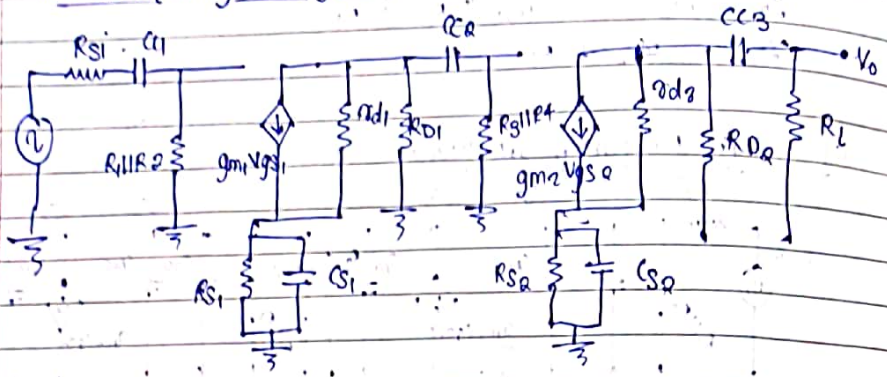
- $f_{Ho_1} = f_{Hi_2}$

  $= \dfrac{1}{2\pi\,(r_{d_1} \| R_3 \| R_4 \| R_{D2})(C_{u_1} + C_{\pi 2} + C_{M_2})}$

- $f_{Ho_2} = \dfrac{1}{2\pi\,(r_{d_2} \| R_L \| R_{D2})\,C_{u_2}}$

- lowest of 3 → f_H

D] Low Frequency Analysis



- $f_{LC1} = \dfrac{1}{2\pi\left[R_{Si} + R_1\|R_2\right] C_{C1}}$

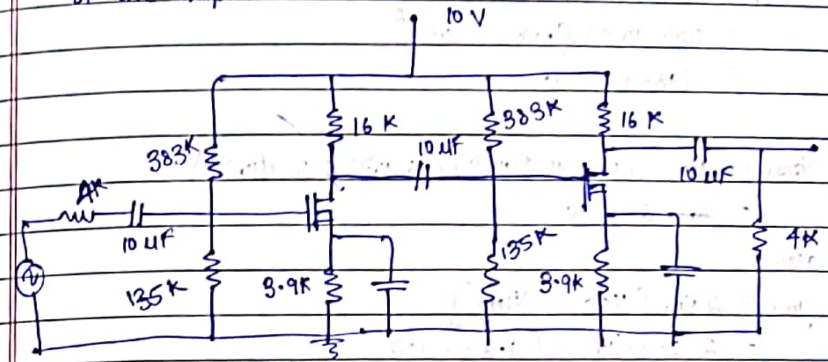
$f_{LC2} = \dfrac{1}{2\pi\left[R_{D1} + R_3\|R_4\right] C_{C2}}$

- $f_{LC3} = \dfrac{1}{2\pi\left[R_{D2} + R_L\right] C_{C3}}$

- $f_{LCS1} = \dfrac{1}{2\pi\left[R_{S1}\|\frac{1}{g_{m1}}\right] C_{S1}}$

$f_{LCS2} = \dfrac{1}{2\pi\left[R_{S2}\|\frac{1}{g_{m2}}\right] C_{S2}}$

9] Design for the given ckt CS-CS MOSFET amp. determine $|A_V|_{mid}$, 3dB f_L, 3dB f_H and BW. of the amplifier.



- K_{n1} = 500 $\mu A/V^2$ $V_{TN1} = V_{TN2} = 1.2$ V
  K_{n2} = 200 $\mu A/V^2$
- $\lambda_1 = \lambda_2 = 0$
- $V_{DD} = 10$ V, $R_{S1} = R_{S2} = 3.9$ K
  $R_1 = 383$ K, $R_2 = 135$ K
  $R_{D1} = 16$ K $= R_{D2}$, $R_{Si} = 4$K
  $R_L = 4$K
- $C_{C1} = C_{C2} = C_{C3} = 10$ uF
- $C_{gs} = 10$ pF, $C_{gd} = 1$ pF
- $C_{S1} = C_{S2} = 100$ uF

→ Solution:

→ For CS (I)

- $V_{Th1} = V_{G1} = \dfrac{V_{DD}(R_2)}{R_1 + R_2}$

$= \dfrac{(10)(135)(10^3)}{518(10^3)} = 2.6$ V

- $V_{GS1} = V_{G1} - I_{D1} R_{S1}$

$= V_{G1} - K_{n1}\left[V_{GS1} - V_{TN1}\right]^2 R_{Si}$

$= 2.6 - (500)(10^{-6})\left[V_{GS1} - 1.2\right]^2 (3.9)(10^3)$

$\therefore V_{GS1} = 2.6 - 1950(10^{-3})\left[V_{GS1}^2 - 2.4V_{GS1} + 1.44\right]$

$= 2.6 - 1.95 V_{GS1}^2 + 4.68 V_{GS1} - 2.8$

$\therefore -1.95 V_{GS1}^2 + 3.68 V_{GS1} - 0.2 = 0$ $V_{GS1} = 1.83$, $V_{GS2} = 0.056$

Now, $V_{GS_1} > V_{TN_1}$

$\therefore V_{GS_1} = 1.83$ V

$\therefore I_{D_1} = K_{n_1}[V_{GS_1} - V_{TN_1}]^2$

$= 500(10^{-6})[1.83 - 1.2]^2$

$= 198.45$ µA

- $V_{DS_1} = 10 - (198.45)(10^{-6})(16(10^3) + 3.9(10^3))$

$= \cancel{10}\ 6.05$ V

also

- $g_{m_1} = 2K_{n_1}[V_{GS_1} - V_{TN_1}]$

$= 2(500)(10^{-6})[1.83 - 1.2]$

$= 0.63$ m℧

→ ~~AC Analysis~~

FOS CS (II) → • $V_{Th_2} = V_{G2} = \dfrac{(10)(135)(10^3)}{518(10^3)}$

$\therefore V_{Th_2} = 2.6$ V

- $V_{GS_2} = V_{G2} - I_{D2} R_{S2}$

$= 2.6 - (200)(10^{-3})[V_{GS_2} - 1.2]^2(3.9)\cancel{(10)}$

$= 2.6 - 0.78[V_{GS_2}^2 - \cancel{2V_{GS}}\ 2.4V_{GS_2} + 1.44]$

$= 2.6 - 0.78V_{GS_2}^2 + 1.872V_{GS_2} - 1.12$

$\therefore -0.78V_{GS_2}^2 + 0.872V_{GS_2} + 1.48 = 0$

$\therefore V_{GS_2} = 2.044$ V

$\therefore I_{D_2} = K_{n_2}[V_{GS_2} - V_{TN_2}]^2$

$= (200)(10^{-6})[2.044 - 1.2]^2$

$= 142.4$ µA

- $V_{DS_2} = 10 - (142.4)(10^{-3})[19.9]$

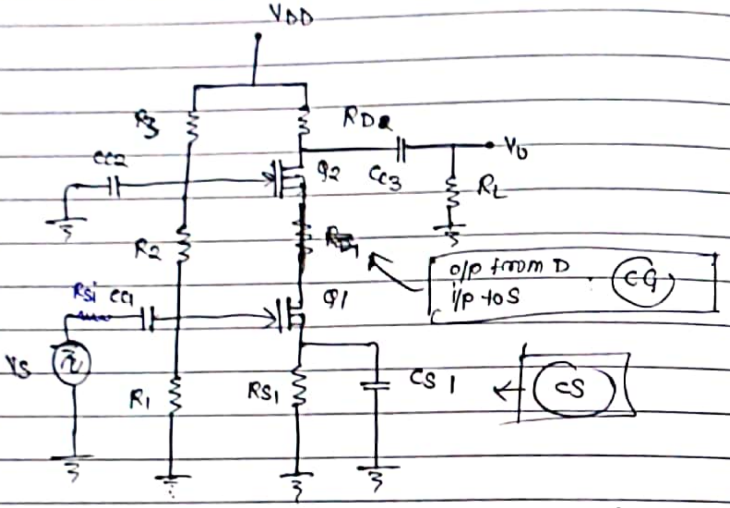
$= 7.166$ V

- $\therefore g_{m_2} = 0.337$ m℧

- $A_{V_1} =$

$\oint \vec{E}\cdot d\vec{s} = q_{enc} = \psi$

* **Cascode Amplifier:** CS-CG MOSFET Amplifier



$Q_1 = (I_{D1}, V_{DS1})$

- Good gain and Good BW. $Q_2 = (I_{D2}, V_{DS2})$

→ DC Analysis:

i) • $V_{GS1} = V_{G1} - I_{D1} R_S$

$V_{G1} = \dfrac{V_{DD} \cdot R_1}{R_1 + R_2 + R_3}$

- $I_{D1} = K_{n1}[V_{GS1} - V_{TN}]^2$, $V_{S1} = I_{D1} \cdot R_S$

$V_{S2} = V_{D1}$

ii) • $V_{GS2} = V_{G2} - V_{S2}$

here,

$V_{G2} = \dfrac{V_{DD}(R_1 + R_2)}{R_1 + R_2 + R_3}$

here, $V_{DS1} = V_{D1} - V_{S1}$

$V_{D2} = V_{DD} - I_{D2} \cancel{R_{S2}} R_D$

$V_{DS2} = V_{D2} - V_{S2}$

$= V_{D2} - V_{D1}$ (here $V_{S2} = V_{D1}$)

$V_{GS2} = V_{G2} - V_{S2}$

∵ Both transistors are identical will be given in question

Q] $V_{DD} = 10$ V, $R_3 = 95$ kΩ, $R_2 = 150$ kΩ, $R_1 = 55$ kΩ

$R_S = 10$ kΩ, $R_D = 2.5$ kΩ, $K_{n1} = 2$ mA/V², $K_{n2} = 1$ mA/V²

$V_{TN1} = V_{TN2} = 0.5$ V

$C_{C1} = C_{C2} = C_{C3} = 10$ µF, $C_S = 100$ µF

→ Solution:

i) $V_{G1} = \dfrac{V_{DD} R_1}{R_1 + R_2 + R_3} = \dfrac{(10)(55)(10^3)}{\cancel{250}\ 300\ \text{k}\Omega} = 1.83$ V

$V_{GS1} = V_{G1} - I_{D1} R_S$

$= V_{G1} - K_{n1}[V_{GS1} - V_{TN}]^2 R_S$

$= 1.83 - (2)(10^{-3})[V_{GS1} - 0.5]^2\ 10(10^3)$

$= 1.83 - 20[V_{GS1}^2 - V_{GS1} + 0.25]$

$= 1.83 - [20 V_{GS1}^2 - 20 V_{GS1} + 5]$

$\therefore V_{GS1} = 1.83 - 20 V_{GS1}^2 + 20 V_{GS1} - 5$

$\therefore 20 V_{GS1}^2 - 19 V_{GS1} + 3.17 = 0$

$\therefore V_{GS1} = 0.73$ V , 0.21

$I_{D1} = K_{n1}[V_{GS1} - V_{TN}]^2$

$= (2)(10^{-3})[0.73 - 0.5]^2$

$= 0.1058$ mA

$V_{S1} = \dfrac{(105.8)(10^{-6})}{(10)(10^3)}$

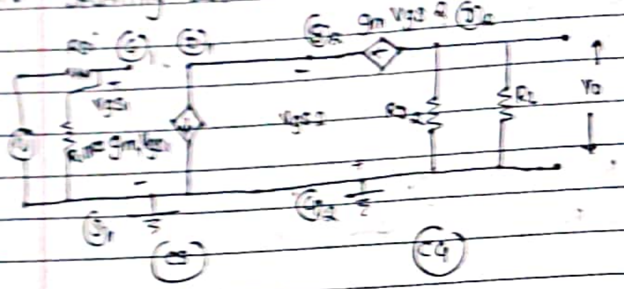
$= 1.058$ V

ii) $V_{G2} = \dfrac{V_{DD}(R_1 + R_2)}{R_1 + R_2 + R_3} = \dfrac{(10)(205)(10^3)}{300(10^3)}$

$= 6.83$ V

→ Mid frequency analysis:

- Starting with lower side:



$\rightarrow$ $Z_i = R_{si} + (R_1 \| R_2)$

$\rightarrow$ $Z_o = R_{D2} \| R_L$

$\rightarrow$ $A_{v1} = \frac{V_o}{V_i} = \frac{-g_{m1} V_{gs1} (1/g_m)}{V_{gs1}} = -1$

$A_{v2} = \frac{V_o}{V_i} = \frac{-g_{m2} V_{gs2} (R_D \| R_L)}{-V_{gs2}} = g_{m2} (R_D \| R_L)$

- Now,

$A_v = A_{v1} \cdot A_{v2}$

$\therefore A_v = -g_{m2} (R_D \| R_L)$

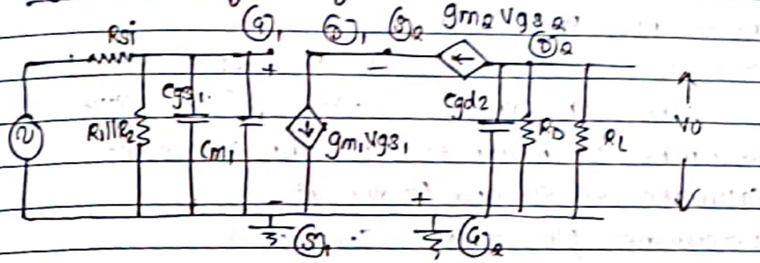
$\because$ both transistors are identical

$\therefore g_{m1} = g_{m2}$

$\therefore A_v = -g_{m1} (R_D \| R_L)$

- The total A_v of cascode amplifier is same as single stage CS amplifier

→ ★ High Frequency Analysis:

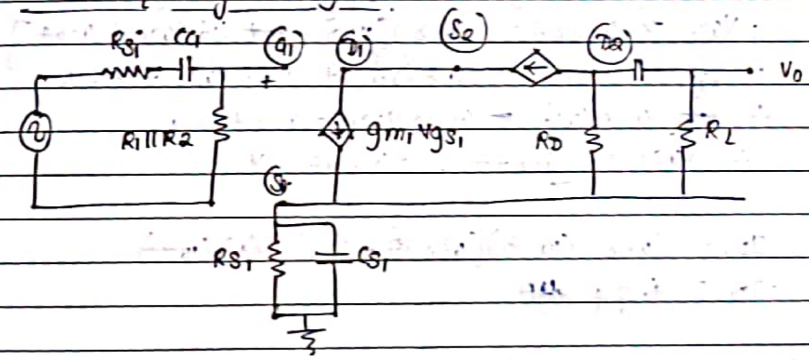


- $F_{Hi} = \frac{1}{2\pi [R_{si} \| R_1 \| R_2](C_{gs1} + C_{m1})}$

$C_{m1} = C_{gd1} [1 + g_{m2} V_{gs2} (R_D \| R_L)]$

$F_{Ho} = \frac{1}{2\pi (R_{D2} \| R_L) C_{gd2}}$

→ Low Frequency Analysis:



- $f_{LC1} = \frac{1}{2\pi (R_{S1} + R_1 \| R_2) C_{C1}}$

$f_{LC2} = \frac{1}{2\pi (R_D + R_L) C_{C2}}$

$f_{LCS1} = \frac{1}{2\pi \left(R_{S1} \| \frac{1}{g_{m1}}\right) C_{S1}}$

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9] For the ckt dgm : $R_1 = 12\,k\Omega$; $R_2 = 24\,k\Omega$, $R_3 = 100\,k\Omega$, $R_S = 1\,k\Omega$, $R_D = 3.3\,k\Omega$, $C_{C1} = C_{C2} = 10\,\mu F$, $C_S = 100\,\mu F$, $V_{DD} = 10\,V$; $R_{Si} = 1\,k\Omega$, $R_L = 10\,k\Omega$, $[K_{n1} = K_{n2} = 0.8\,mA/V^2]$, $[V_{TN1} = V_{TN2} = 0.8\,V]$

Determine DC parameters of given cascode ckt,
Calculate Z_i, Z_o and A_v for ckt.
Determine BW of ckt. ($I_{D1} = I_{D2}$)

→ Solution:

A) DC Analysis:

$$V_{G1} = \frac{V_{DD} \cdot R_1}{R_1 + R_2 + R_3} = \frac{(10)(12)(10^3)}{136\,(10^3)} = 0.882\ V$$

Now, $V_{GS1} = V_{G1} - I_{D1} R_S$

$$= V_{G1} - K_{n1}[V_{GS1} - V_{TN1}]^2 R_S$$

$$= 0.882 - (0.8)(10^{-3})[V_{GS1} - 0.8]^2 (1)(10^3)$$

$$\therefore V_{GS1} = 0.882 - (0.8)[V_{GS1}^2 - \cancel{2} 1.6\,V_{GS1} + 0.64]$$

$$\therefore V_{GS1} = 0.882 - 0.8V_{GS1}^2 + 1.28\,V_{GS1} - 0.512$$

$$\therefore \cancel{+8}\ 0.8V_{GS1}^2 - 0.28\,V_{GS1} - 0.37 = 0$$

$$\therefore V_{GS1} = \underline{0.877},\ -0.527\ V$$

$$I_{D1} = (0.8)(10^{-3})(0.877 - 0.8)^2 \cancel{(1)(10^3)}$$

$$\therefore I_{D1} = 4.74\ \cancel{mA}$$

$$V_{G2} = \frac{V_{DD}(R_1 + R_2)}{136\,(10^3)} = \frac{360\,(10^3)}{136\,(10^3)} = 2.647\ V$$

Now, $V_{GS2} = V_{G2} - I_{D2}(R_S)$
$= 2.64 - (4.74)(10^{-3})(1)$
$= 2.63$

→ $V_{D2} = V_{DD} - I_{D2} \cdot R_D$

$$= 10 - (4.74)(10^{-6})(3.3)(10^3)$$

$$= 9.99\ V$$

$$V_{S1} = I_{D1} \cdot R_S$$

$$= (4.74)(10^{-6})(1)(10^3)$$

$$= 4.74\ mV$$

Now, $V_{S2} = V_{D1}$
Now,
$V_{DS2} = \quad V$

→ Thus, ∵ $I_{D1} = I_{D2}$ and $V_{TN1} = V_{TN2}$ & $K_{n1} = K_{n2}$

$$\therefore V_{GS1} = V_{GS2} = 0.877\ V$$

B) Now,

$$g_m = 2K_n(V_{GS} - V_{TN})$$

$$= 2(0.8)(10^{-3})(0.877 - 0.8)$$

$$= 0.1232\ mV$$

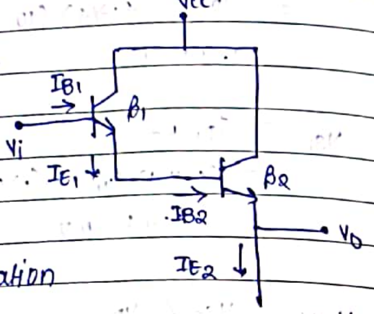
$$\therefore |A_v|_{mid} = 0.1232\,(10^{-3})(3.3\,k \parallel 10\,k)$$

$$= 0.305\ \cancel{(10^{-3})}$$

C)

$$f_{L1} = \frac{1}{2\pi\,(R_{Si} + R_1 \parallel R_2)\,C_{C1}}$$

* Darlington Amplifier : (CC - CC pair)



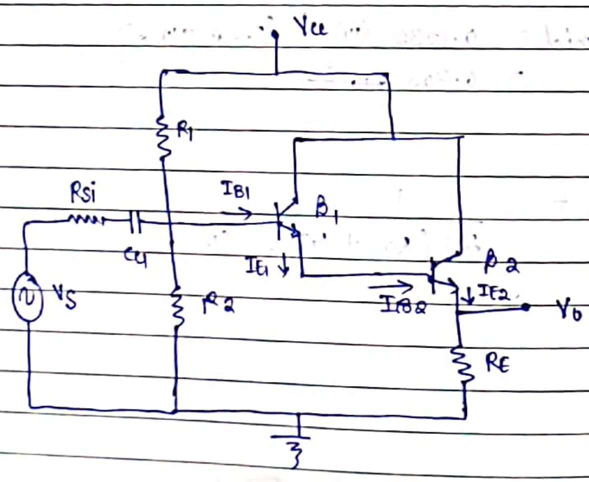
- used to improve current gain in amplifier. It has high i/p Z and low o/p Z
- It can be used as a buffer amplifier for isolation purpose.
- Not used for v-amplification, coz. Av for CC is always 1

- $A_i = \frac{I_{E2}}{I_{B1}} = \frac{(1+\beta_2) I_{B2}}{I_{B1}}$

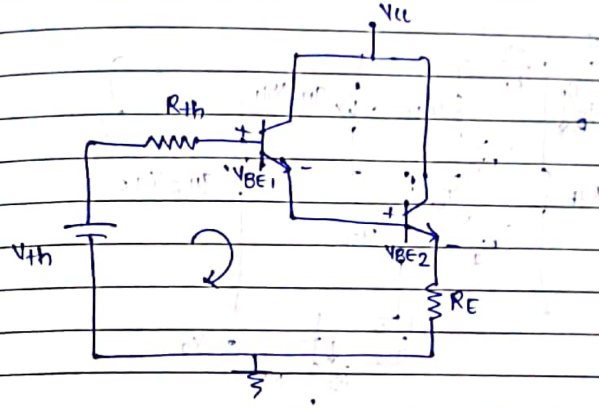
But, $I_{B2} = I_{E1} = (1+\beta_1) I_{B1}$

$\therefore A_i = \frac{(1+\beta_2)(1+\beta_1) I_{B1}}{I_{B1}}$

$\therefore A_i = (1+\beta_2)(1+\beta_1)$
$\approx \beta_1 \cdot \beta_2$
$\approx \beta^2$

→


→ DC equivalent ckt :



- $V_{th} = \frac{V_{cc} \cdot R_2}{R_1 + R_2}$, $R_{th} = R_1 \parallel R_2$

- Now,
$V_{th} - I_{B1} \cdot R_{th} - V_{BE1} - V_{BE2} - I_{E2} \cdot R_E = 0$
$\boxed{\therefore I_{B1} = \frac{V_{th} - V_{BE1} - V_{BE2}}{R_{th} + \beta_D \cdot R_E}}$

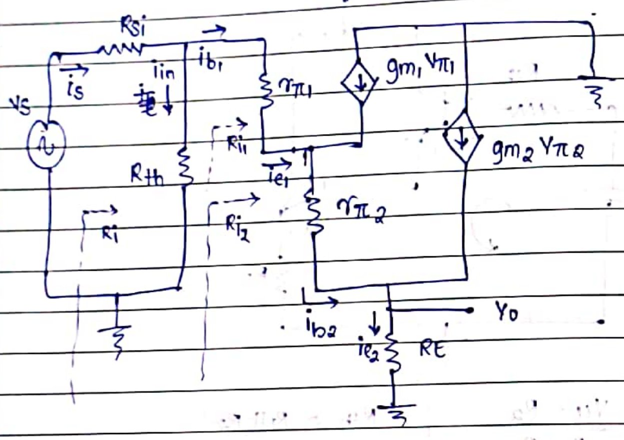
here,
- $I_{E2} = (1+\beta_2) I_{B2}$
$= (1+\beta_2) I_{E1}$
$= (1+\beta_2)(1+\beta_1) I_{B1}$
$\approx \beta_1 \beta_2 \; I_{B1}$
$\therefore I_{E2} \approx \beta_D \cdot I_{B1}$

- $\boxed{I_{C1} = \beta_1 I_{B1}}$

- Now, $V_{cc} - V_{CE2} - I_{E2} R_E = 0$
$\boxed{\therefore V_{CEQ} = V_{cc} - I_{EQ} R_E}$

$I_{E1} = I_{C1} + I_{B1}$
$I_{C2} = \beta_2 I_{B2}$
$\boxed{V_{E2} = I_{E2} R_E}$

→ Hybrid - π model



→ Total i/p R :

- $R_{in} = R_{si} + R_i$
- Now, $R_i = R_{th} \parallel R_{i_1}$
- here, $R_{i_1} = r_{\pi_1} + (1+\beta_1)\left[r_{\pi_2} + (1+\beta_2)R_E\right]$

Thus,

⇒ • $R_{in} = R_{si} + \left[R_{th} \parallel \left(r_{\pi_1} + (1+\beta_1)\left[r_{\pi_2} + (1+\beta_2)R_E\right]\right)\right]$

→ o/p R :

- $R_o = R_E \parallel R_{o_1}$

Now, $R_{o_1} = \dfrac{R_{o_2} + r_{\pi_2}}{1+\beta_2}$

here, $R_{o_2} = \cancel{R_{si} \parallel R_1 \parallel R_2}\ \dfrac{R_{si} \parallel (R_{th} + r_\pi)}{1+\beta_1}$

⇒ • $R_o = R_E \parallel \left[\dfrac{\left(\dfrac{R_{si} \parallel (R_{th} + r_\pi)}{1+\beta_1}\right) + r_{\pi_2}}{1+\beta_2}\right]$

→ Current gain :

~~Ai =~~ $A_i = \dfrac{i_{e_2}}{i_s}$

Now, $A_i = \dfrac{i_{e_2}}{i_{b_2}} \cdot \dfrac{i_{b_2}}{i_{e_1}} \cdot \dfrac{i_{e_1}}{i_{b_1}} \cdot \dfrac{i_{b_1}}{i_{in}} \cdot \dfrac{i_{in}}{i_s}$

$= (1+\beta_2)(1)(1+\beta_1)\left[\dfrac{R_{th}}{R_{th} + R_{i_1}}\right]\left[\dfrac{R_{si}}{R_{si} + R_i}\right]$

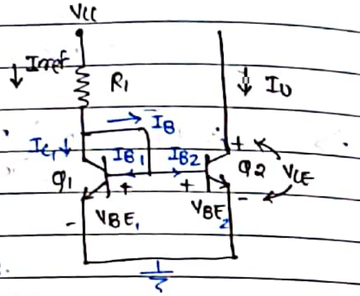
⇒ $\therefore A_i = \beta_D \left[\dfrac{R_{th}}{R_{th} + R_{i_1}}\right]\left[\dfrac{R_{si}}{R_{si} + R_i}\right]$

Module 2

Integrated ckt biasing techniques

* const. current source biasing :

(A) 2 transistor const. current source (current Mirror) :



- E of both qs are tied together and collector of both qs connected to base, ∴ q_1 will act as a diode.
- As V_{cc} is applied to q_1, I_{ref} will flow making q_1 - FB. and as q_1 gets FB, q_2 will also get FB. and I_o flows through q_2 line
- KVL on i/p side

$$\therefore V_{cc} - I_{ref} \cdot R_1 - V_{BE_1} = 0$$

$$\therefore I_{ref} = \frac{V_{cc} - V_{BE_1}}{R_1}$$

- Also, $I_{ref} = I_{c_1} + I_B$
$= I_{c_1} + I_{B_1} + I_{B_2}$

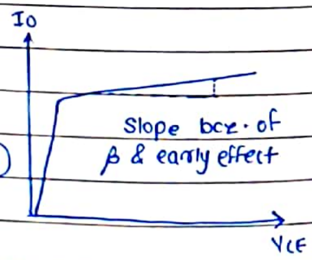
∵ Bases are tied together ⇒ ∴ $I_{B_1} = I_{B_2}$

$$\therefore I_{ref} = I_{c_1} + 2I_{B_2} = I_{c_1} + \frac{2I_{c_2}}{\beta}$$

∵ q_1 & q_2 are identical ∴ $I_{c_1} = I_{c_2}$

$$\therefore I_{ref} = I_{c_2} + \frac{2I_{c_2}}{\beta} = I_{c_2}\left[1+\frac{2}{\beta}\right]$$

$$\therefore I_{c_2} = I_o = \frac{I_{ref}}{\left[1+\frac{2}{\beta}\right]} \quad \text{—(I)}$$



$$\therefore \frac{I_o}{I_{ref}} = \frac{1}{\left[1+\frac{2}{\beta}\right]} \cdot \frac{\left(1+\frac{V_{CE2}}{V_A}\right)}{\left(1+\frac{V_{CE1}}{V_A}\right)}$$

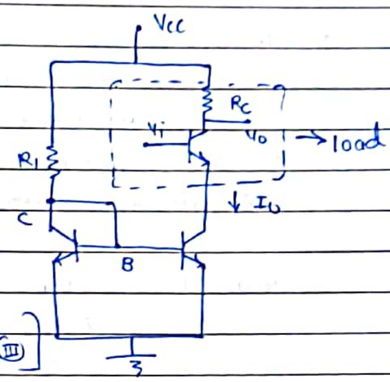
here V_A : early v.

Example

- Now, from adj dgm.

→ $V_{CE_2} = V_i - V_{BE}$ —(II)

→ $V_{CE_1} = V_{BE}$ —(III)



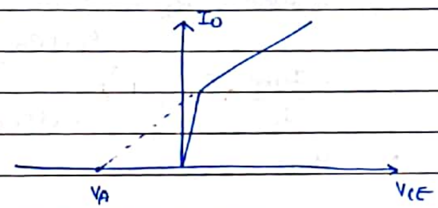
Now,

$$\frac{dI_o}{dV_{CE_2}} = \frac{I_{ref}}{\left[1+\frac{2}{\beta}\right]} \cdot \frac{\left(1/V_A\right)}{\left[1+\frac{V_{BE}}{V_A}\right]}$$

$$\therefore \frac{1}{r_o} = \frac{I_{ref}}{\left[1+\frac{2}{\beta}\right]} \cdot \frac{1}{V_A}$$ → [From (I),(II),(III)]

$$\therefore \frac{1}{r_o} = \frac{I_o}{V_A}$$

$$\therefore r_o = \frac{V_A}{I_o}$$



→ $r_o : \infty$ to achieve const. current source

→ Independent of β

→ condns. to get good const. current source

Q] Describe 2 q - const. i source & derive relationship betn. I_o & I_{ref}

Q] Explain the effect of r_o in const. i source

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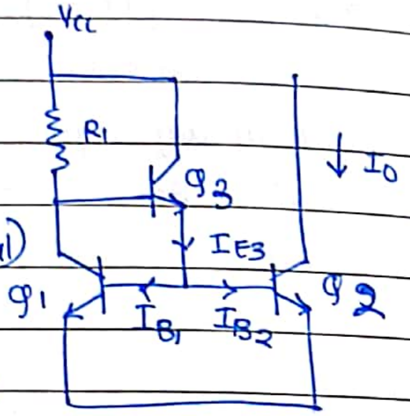
(B) Basic 3 transistor current source:

- q_3 supplies base current I_{B_1} & I_{B_2} to q_1 & q_2 resp.



- Here, $I_{B_1} = I_{B_2}$

$I_{C_1} = I_{C_2}$ ($\because q_1, q_2$ identical)

$I_{Ref} = I_{C_1} + I_{B_3}$

→ $I_{E_3} = I_{B_1} + I_{B_2}$ (KCL at q_3)

$\therefore (1+\beta) I_{B_3} = 2 I_{B_2}$

Now, $I_{ref} = I_{C_1} + I_{B_3}$

$$= I_{C_1} + \frac{I_{E_3}}{1+\beta_3} = I_{C_1} + \frac{2 I_{B_2}}{1+\beta_3}$$

$$\therefore I_{ref} = I_{C_2} + \frac{2 I_{C_2}}{\beta(1+\beta_3)}$$

$$\therefore \frac{I_{ref}}{I_{C_2}} = \left[1 + \frac{2}{\beta(1+\beta_3)}\right]$$

- thus, $I_{C_2} = I_o = \dfrac{I_{ref}}{\left[1 + \frac{2}{\beta(1+\beta_3)}\right]}$

→ here, in improved version of 3-q current source, change in load current with change in β is much smaller than 2-q current source.

- $I_{ref} = \dfrac{V_{CC} - V_{BE_3} - V_{BE}}{R_1}$